Python Fundamentals II

- String Manipulation, Input & Output,
- Branching and Iteration

Lecture Objectives

- - Understand strings and string operations
- - Use input and output functions
- - Apply decision making (branching)
- - Use loops for repetition

Strings in Python

- - A string is a sequence of characters
- - Defined using single or double quotes
- - Example: name = "Python"

String Operations

- - Indexing and slicing
- - Concatenation (+)
- - Repetition (*)
- - Length using len()

Common String Methods

- - upper(), lower()
- - strip()
- - replace()
- - split()

Input in Python

- - input() collects user data
- - Input is always a string
- - Use type casting when necessary

Output in Python

- - print() displays output
- - Formatted output using f-strings

Branching (Decision Making)

- - if statement
- - if...else statement
- - if...elif...else

Comparison Operators

- > , < , == , != , >= , <=

Logical Operators

- and , or , not

Iteration (Loops)

- - for loop
- - while loop

Loop Examples

- - Looping through range
- - Looping through lists

Statistics Application

- - Sum of values
- - Average calculation
- - Score classification

Summary

- - Strings handle text data
- - Input & Output enable interaction
- - Branching makes decisions
- - Iteration automates repetition